

Efficacy and safety of once-daily oral orforglipron compared with oral semaglutide in adults with type 2 diabetes (ACHIEVE-3): a multinational, multicentre, non-inferiority, open-label, randomised, phase 3 trial



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Summary

Background Orforglipron is a novel non-peptide (GLP-1) receptor agonist designed for daily oral administration without food or water restrictions. This study aimed to compare the efficacy and safety of orforglipron with oral semaglutide in individuals with type 2 diabetes inadequately controlled with metformin.

Methods In this 52-week, randomised, open-label, active-controlled, multicentre, multinational, phase 3 study, we enrolled adults (≥ 18 years) with type 2 diabetes inadequately controlled with metformin (≥ 1500 mg per day), glycated haemoglobin (HbA_{1c}) between 7·0% and 10·5% (53–91 mmol/mol), and BMI at least 25 kg/m² from 131 medical research centres and hospitals in Argentina, China, Japan, Mexico, and the USA. Participants were randomly assigned (1:1:1:1) to orforglipron (12 mg or 36 mg) or semaglutide (7 mg or 14 mg); all groups had an up to 4-week lead-in period and 52-week treatment period, with the drugs administered orally once per day. The primary objective of the study was to assess non-inferiority of orforglipron 36 mg versus semaglutide 14 mg and orforglipron 12 mg versus semaglutide 7 mg for mean change at week 52 from baseline in HbA_{1c} (with a non-inferiority margin of 0·3%) in the intention-to-treat population. Hierarchical analysis for superiority was prespecified after attainment of non-inferiority. The treatment regimen estimand, based on data from all randomly assigned participants regardless of intercurrent events, was the primary estimand; the efficacy estimand was considered supportive. The safety endpoints used data from all participants who received at least one dose of the study drug. This trial was registered on ClinicalTrials.gov (NCT06045221) and is completed.

Findings From Sept 22, 2023, to Aug 22, 2025, 1698 participants were recruited and randomly assigned to orforglipron (n=424 on 12 mg and n=423 on 36 mg) or semaglutide (n=426 on 7 mg and n=425 on 14 mg). For the treatment regimen estimand, mean changes at week 52 from a baseline HbA_{1c} of 8·3% were –1·71% (SE 0·07) with orforglipron 12 mg, –1·91% (0·08) with orforglipron 36 mg, –1·23% (0·05) with semaglutide 7 mg, and –1·47% (0·06) with semaglutide 14 mg. Estimated treatment differences were –0·48% (95% CI –0·65 to –0·31; p<0·0001) for orforglipron 12 mg versus semaglutide 7 mg; –0·44% (–0·62 to –0·26; p<0·0001) for orforglipron 36 mg versus semaglutide 14 mg; –0·24% (95% CI –0·41 to –0·072; p=0·0050) for orforglipron 12 mg versus semaglutide 14 mg; and –0·68% (–0·85 to –0·50; p<0·0001) for orforglipron 36 mg versus semaglutide 7 mg. The primary objective of non-inferiority was met and both orforglipron doses showed superiority to both semaglutide doses, including orforglipron 12 mg versus semaglutide 14 mg. The most frequent adverse events were gastrointestinal events (orforglipron: 249 [59%] of 424 on 12 mg and 245 [58%] of 423 on 36 mg; semaglutide: 157 [37%] of 426 on 7 mg and 193 [45%] of 425 on 14 mg), most of which were mild to moderate in severity. More participants in the orforglipron groups (37 [9%] on 12 mg and 41 [10%] on 36 mg) discontinued study treatment due to adverse events than in the semaglutide groups (19 [4%] on 7 mg and 21 [5%] on 14 mg), and mean increase in pulse rate was greater in the orforglipron groups (12 mg 3·7 bpm; 36 mg 4·7 bpm) than in the semaglutide groups (7 mg 1·0 bpm; 14 mg 1·5 bpm). Four deaths occurred during the study: one in the orforglipron 12 mg group, one in the orforglipron 36 mg group, and two in the semaglutide 7 mg group.

Interpretation In individuals with type 2 diabetes inadequately controlled with metformin, orforglipron 12 mg and 36 mg was non-inferior and superior to semaglutide 7 mg and 14 mg with respect to the mean change in HbA_{1c} from baseline to 52 weeks. Although the safety profiles of both orforglipron and semaglutide were generally consistent with the GLP-1 receptor agonist class, the incidence of gastrointestinal events, discontinuations due to adverse events, and mean increase in pulse rate were higher with orforglipron than oral semaglutide.

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See Online for appendix

Research in context

Evidence before this study

GLP-1 receptor agonists have become a cornerstone in the management of type 2 diabetes, primarily due to their efficacy in improving glycaemic control and weight reduction. On Nov 20, 2025, we searched PubMed with the search terms “glucagon-like peptide-1 receptor agonist”, AND “oral”, AND “type 2 diabetes”, AND “overweight” and limited our search to clinical trials, with no date or language restrictions. Most approved GLP-1 receptor agonists are peptide-based and require subcutaneous administration, with oral semaglutide, co-formulated with an absorption enhancer, sodium N-(8-[2-hydroxybenzoyl]amino) caprylate, being the only oral option. Oral semaglutide has restrictions on food and fluid intake to improve bioavailability. Phase 2 clinical trials showed that orforglipron, a non-peptide, orally available, small molecule GLP-1 receptor agonist, could achieve dose-dependent reductions in glycated haemoglobin (HbA_{1c}) and bodyweight over 26 weeks. In addition to type 2 diabetes and an ongoing cardiovascular outcomes trial, orforglipron is under development for obesity (NCT05869903 and NCT05872620), obstructive sleep apnoea (NCT06649045), stress urinary incontinence (NCT07202884), and peripheral artery disease (NCT07223593). ACHIEVE-1, the first phase 3 clinical trial of orforglipron, showed clinically meaningful efficacy for reduction in HbA_{1c} and bodyweight and safety consistent with the GLP-1 receptor agonist class in individuals with early type 2 diabetes inadequately controlled with diet and exercise alone. In individuals with type 2 diabetes and obesity or overweight (ATTAIN-2), orforglipron was superior for HbA_{1c} and weight reduction than placebo. For individuals or health-care providers considering initiating an oral GLP-1 receptor agonist, there is no evidence on the comparative efficacy and safety of orforglipron versus oral semaglutide.

Added value of this study

ACHIEVE-3 is a multicentre, open-label, international, randomised, active-controlled phase 3 trial to directly compare orforglipron (12 mg or 36 mg) with oral semaglutide (7 or 14 mg) in adults with type 2 diabetes inadequately controlled with metformin. The findings show that orforglipron was non-inferior and superior to semaglutide for HbA_{1c} reduction. Orforglipron 12 mg was significantly better than semaglutide 7 mg and orforglipron 36 mg was significantly better than both semaglutide 7 mg and 14 mg for achieving glycaemic targets and robust weight reduction. Additionally, orforglipron 12 mg was superior to semaglutide 14 mg for HbA_{1c} change from baseline and the proportion of participants achieving HbA_{1c} of less than 7.0% and 6.5% or less. The frequency of gastrointestinal adverse events and treatment discontinuation due to gastrointestinal adverse events, and the increase from baseline in pulse rate, were higher in orforglipron groups than semaglutide groups. The overall safety profiles of both orforglipron and semaglutide were overall consistent with the GLP-1 receptor agonist class.

Implications of all the available evidence

Together with previous research, the results of the ACHIEVE-3 study suggest that orforglipron represents an important advancement in the oral treatment landscape for type 2 diabetes. Its efficacy, safety, tolerability, and simple dosing could address important barriers associated with current incretin-based therapies, offering a new highly efficacious and safe option for individuals seeking effective glycaemic and weight control without the use of injections or administration restrictions.

Introduction

GLP-1 receptor agonists have gained increasing prominence in the management of type 2 diabetes due to their pleiotropic benefits, including glycaemic control, weight reduction, and cardiorenal risk reduction.^{1–3} These therapies act by stimulating insulin secretion and suppressing glucagon secretion, both in a glucose-dependent manner, decreasing appetite, and reducing bodyweight. Additionally, observed benefits include reductions in blood pressure,⁴ reductions in visceral adiposity,⁵ and improvements in lipid profiles,⁶ particularly diabetic dyslipidaemia.⁷

Most GLP-1 receptor agonists are peptide-based, with a large molecular weight.⁸ Consequently, these therapies are currently administered via subcutaneous injection to allow absorption, avoid proteolytic degradation in the gastrointestinal tract, and minimise the effect of first-pass metabolism. However, some individuals are hesitant to initiate injectable therapies, which can delay timely treatment initiation and intensification.^{9,10} An oral non-peptide GLP-1 receptor agonist with the physiological

benefits of peptide-based GLP-1 receptor agonists, without the need for injections, would allow further tailoring of therapy to meet individual preferences of patients and prescribers.

Oral semaglutide is a peptide-based GLP-1 receptor agonist co-formulated with an absorption enhancer, sodium N-(8-[2-hydroxybenzoyl]amino) caprylate. Due to the low bioavailability of this formulation (~1–2%),¹¹ to achieve therapeutic concentrations, administration must occur in a fasted state in the morning, with no more than 120 mL of water, and at least 30 min before food, beverages, or other oral medication.^{12,13} Oral semaglutide is currently the only orally administered GLP-1 receptor agonist approved for the management of type 2 diabetes, with additional indications for risk reductions in cardiovascular and chronic kidney diseases.^{14–16}

Orforglipron is a non-peptide (small molecule), partial GLP-1 receptor agonist biased towards G-protein activation versus β -arrestin recruitment, with pharmacokinetic characteristics favourable for once per day oral dosing without dietary restrictions.⁸ Orforglipron is under

investigation for several potential indications, including treatment of type 2 diabetes and obesity.^{17,18} In the 40-week phase 3 ACHIEVE-1 study, under the treatment regimen estimand, orforglipron as a monotherapy reduced HbA_{1c} by up to 1·5% (1·1% treatment difference *vs* placebo) from a mean overall baseline of 8·0% and reduced bodyweight by up to 7·6% (5·9% treatment difference *vs* placebo) from a mean overall baseline of 90·2 kg in people with early type 2 diabetes inadequately controlled with diet and exercise.¹⁹

There are no published randomised controlled trials addressing the comparative efficacy and safety of oral GLP-1 receptor agonist therapies. We aim to address this evidence gap by investigating the non-inferiority and superiority of orforglipron (12 mg and 36 mg) compared with oral semaglutide (7 mg and 14 mg) in participants with type 2 diabetes who had inadequate glycaemic control with metformin monotherapy.

Methods

Study design and participants

This 52-week, open-label, randomised, active comparator controlled, parallel-arm, non-inferiority, phase 3 study (ACHIEVE-3) was done in 131 medical research centres and hospitals in Argentina, China, Japan, Mexico, and the USA.

Key inclusion criteria were being aged 18 years or older with type 2 diabetes inadequately controlled with metformin at a dose of at least 1500 mg per day, with HbA_{1c} of 7·0% (53 mmol/mol) to 10·5% (91 mmol/mol), a BMI of 25 kg/m² or more, and a stable bodyweight ($\leq$ 5% bodyweight gain or loss) during the 3 months before screening.

Key exclusion criteria were a history of severe hypoglycaemia within 6 months of screening, type 1 diabetes, current or planned treatment for diabetic retinopathy and/or macular oedema, estimated glomerular filtration rate of less than 45 mL per min per 1·73 m², New York Heart Association Functional Classification IV congestive heart failure, a family or personal history of medullary thyroid carcinoma or multiple endocrine neoplasia syndrome type 2, any liver disease (except metabolic dysfunction-associated steatotic liver disease), AST and/or ALT concentrations more than 5 times upper limit of normal, and a history of chronic or acute pancreatitis. A complete list of eligibility criteria is available in the appendix (pp 5–8).

The trial was registered on ClinicalTrials.gov (NCT06045221) and is complete. The trial protocol was approved by local institutional review boards at all participating sites (appendix p 14). The trial was done in accordance with the principles of the Declaration of Helsinki and the Good Clinical Practice guidelines of the International Council for Harmonisation. All participants provided written informed consent. The statistical analysis plan is available with the protocol in the appendix.

Randomisation and masking

Participants were randomly assigned (1:1:1:1) to receive either orforglipron (escalated up to doses of 12 mg or 36 mg) or oral semaglutide (escalated up to doses of 7 mg or 14 mg) through an interactive web-response system and stratified by country and HbA_{1c} ($\leq$ 8·0% or $>$ 8·0%) at visit 1. Treatment assignment was open label; however, all investigators (including those assessing data), participants, and the sponsor were masked to the randomly assigned dose of orforglipron. Orforglipron capsules were manufactured visually indistinguishable and packaged in identical bottles, distinguished only by a unique package number to support masked dispensing at clinical sites.

Procedures

The study period included a screening and lead-in period of up to 4 weeks, a 52-week treatment period, and a safety follow-up period of 2–5 weeks. Genetic sex (male or female), race, and ethnicity were collected based on participant self-report.

Participants receiving orforglipron started with 1 mg daily, increasing every 4 weeks until reaching their randomly assigned dose of 12 mg or 36 mg. Those receiving semaglutide began with 3 mg daily, taken at least 30 min before the first food, drink, or other oral medications of the day with up to 4 ounces (120 mL) of plain water, and had increasing doses every 4 weeks until reaching their randomly assigned dose of 7 mg or 14 mg (appendix p 21). Initiation of new glucose-lowering medications was allowed according to specific criteria for rescue therapy for severe persistent hyperglycaemia (appendix pp 8–9). Other GLP-1 receptor agonists, DPP4 inhibitors, or other incretin-based therapies were not allowed. In case of intolerable gastrointestinal symptoms, participants were advised to have smaller meals, use symptomatic medications and, if needed, temporarily interrupt use of the study drug for up to two consecutive doses for orforglipron or up to seven consecutive doses for semaglutide (equivalent to one half-life). If intolerable gastrointestinal symptoms persisted, de-escalation of the study drug to the previous lower dose for at least 8 weeks could be considered for participants taking orforglipron 6 mg and more or semaglutide 14 mg. Only one dose de-escalation and re-escalation attempt was permitted during the study. All participants continued on stable metformin as a background glucose-lowering medication throughout the study, except for those with safety concerns.

Participants who prematurely discontinued the trial treatment were encouraged to stay in the study and complete the scheduled visits and assessments.

Ophthalmological evaluation of fundus photographs for diabetic retinopathy or macular oedema (taken by an ophthalmologist or optometrist or at the local clinic) was done for all participants at baseline and week 52, with an additional assessment at week 24 for those found to have

any degree of diabetic retinopathy at baseline. Further details of fundus photography assessments are provided in the appendix (pp 9–10).

Clinic visits occurred every 4 weeks for the first 24 weeks, followed by visits at weeks 32, 40, and 52. Details of the schedule of efficacy and safety assessments are provided in the protocol available at the end of the appendix.

Outcomes

The primary endpoint was the change in HbA_{1c} concentration from baseline to week 52. The key secondary endpoints assessed at week 52 were the proportion of participants with an HbA_{1c} concentration of less than 7·0% (53 mmol/mol) and 6·5% or less (48 mmol/mol), the percentage change from baseline in bodyweight, and the absolute change from baseline in bodyweight. Other secondary endpoints at week 52 were the proportion of participants with an HbA_{1c} concentration of less than 5·7% (39 mmol/mol), the proportion with a weight loss of at least 5%, at least 10%, and at least 15%, the change from baseline in fasting serum glucose, the change from baseline in daily mean 7-point self-monitored blood glucose, percentage change from baseline in lipid measures, and the change from baseline in waist circumference, BMI, and systolic and diastolic blood pressure. Patient-reported outcomes collected during the study were Short Form–36 Health Survey Version 2, EQ-5D-5L, Ability to Perform Physical Activities of Daily Living, Impact of Weight on Self-Perception, Impact of Weight on Quality of Life-Lite Clinical Trials Version, Diabetes Treatment Satisfaction Questionnaire—Status, Diabetes Treatment Satisfaction Questionnaire—Change, and Simplicity of Diabetes Treatment Questionnaire. Only the Short Form–36 Health Survey Version 2 is reported here; other patient-reported outcomes will be published in a programme-level health-outcomes-focused manuscript.

Safety endpoints were the number and incidence of serious adverse events, treatment-emergent adverse events, and discontinuations due to adverse events. Adverse events were coded with the Medical Dictionary for Regulatory Activities and assessed for severity and relationship to study treatment by the study investigators. Diabetic retinopathy was monitored through ophthalmological evaluation of fundus photographs. Physical examinations, vital sign measurements, electrocardiograms, and clinical laboratory assessments were done as outlined in the protocol. Hepatic safety was evaluated with liver function tests at every clinic visit. Hypoglycaemic episodes were recorded in the study eDiary and reported as a serious adverse event if the event met the criteria for severe hypoglycaemia (altered mental or physical status requiring assistance from another person to administer carbohydrates, glucagon, or other resuscitation measures). Additionally, deaths, major adverse cardiovascular events (eg, myocardial

infarction, hospitalisation for unstable angina or heart failure, coronary interventions, and cerebrovascular events), and potential pancreatitis cases were reviewed by an independent external adjudication committee.

Statistical analysis

We estimated that 1576 participants would provide at least 90% power to assess the non-inferiority of orforglipron 36 mg to semaglutide 14 mg and of orforglipron 12 mg to semaglutide 7 mg at the two-sided 0·025 significance level (0·05 overall α split between comparisons) with a two-sample *t* test based on the primary endpoint (ie, change from baseline in HbA_{1c} at week 52), with a non-inferiority margin of 0·3%, a common SD of 1·2%, and assuming no expected difference between orforglipron 12 mg and semaglutide 7 mg or between orforglipron 36 mg and semaglutide 14 mg under the treatment regimen estimand. Additionally, the sample size ensured 90% power to assess superiority of orforglipron 36 mg over semaglutide 14 mg and of orforglipron 12 mg over semaglutide 7 mg, assuming a 0·3% greater reduction in HbA_{1c}. These assumptions also ensured the power for the efficacy estimand. We did power calculations with nQuery (version 9.1.0.0).

We did efficacy analyses in all randomly assigned participants. Two estimands were prespecified: the treatment regimen estimand and the efficacy estimand. For each estimand, we applied a graphical testing procedure to control multiplicity, strongly controlling the overall family-wise (type 1) error rate at 0·05 for the primary and key secondary endpoints, with an initial $\alpha=0\cdot025$ equally allocated to each of the two primary hypotheses: orforglipron 36 mg versus semaglutide 14 mg and orforglipron 12 mg versus semaglutide 7 mg (appendix p 22). Inferences regarding treatment effects in this Article pertain primarily to the treatment regimen estimand.

The treatment regimen estimand assessed the treatment effect between orforglipron and semaglutide, irrespective of premature discontinuation or the use of additional antihyperglycaemic medication (ie, in the intention-to-treat population). We analysed continuous outcomes at week 52 with an ANCOVA model adjusted for baseline value, country, and the stratification factor (baseline HbA_{1c} $\leq 8\cdot0\%$ or $>8\cdot0\%$), and interactions of treatment-by-baseline and treatment-by-stratification factor, incorporating imputed data for missing values at baseline and missing endpoints at week 52, as detailed in the statistical analysis plan. The efficacy estimand was used to assess the treatment effect between orforglipron and semaglutide under the assumption that all the participants who were randomly assigned continued to receive orforglipron or semaglutide as assigned for 52 weeks without the use of additional glucose-lowering medications for more than 14 days. For the efficacy estimand, we used a restricted maximum likelihood-based mixed model (MMRM) for repeated measures to analyse the continuous outcomes. The MMRM included fixed effects for treatment, visit,

country, other strata, baseline value of the dependent variable, baseline-by-treatment-by-visit interaction, and strata-by-treatment-by-visit interaction. Under both estimands, we analysed binary efficacy outcomes with the use of logistic regression; we addressed missing values through estimand-specific multiple imputation strategies, with results combined according to Rubin's rule.²⁰ We did prespecified subgroup analyses for the change from baseline in HbA_{1c} at week 52 by baseline HbA_{1c} stratum ($\leq 8.0\%$ and $> 8.0\%$).

We assessed safety endpoints with data from the safety population (all participants who were randomly assigned to a study intervention and received at least one dose of the study intervention) during treatment and the safety follow-up period, regardless of adherence. Assessment of hypoglycaemia excluded events that occurred after initiation of any new antihyperglycaemic therapy. Additional details regarding the statistical analysis are provided in the appendix (pp 10–13).

The frequency and proportion of participants experiencing adverse events and serious adverse events were summarised descriptively. We used Fisher's exact test for categorical comparisons and provide risk differences with 95% CIs. We analysed some prespecified

continuous measures with MMRM or ANCOVA with relevant covariates. Analyses of safety endpoints were exploratory. No explicit imputations were done for safety measures.

No dedicated data monitoring committee was used. An unmasked programme-level committee regularly reviewed safety data from ACHIEVE-4, the longest orforglipron study in participants with high cardiovascular risk. This committee could assess data from other studies, including ACHIEVE-3, if routine masked monitoring of adverse events by the sponsor identified a need, or if the committee deemed this was warranted based on open-label review of ACHIEVE-4 data, but no such need arose during ACHIEVE-3 or other studies.

Statistical analyses were done with SAS (version 9.4) and R (version 3.5 and later).

Role of the funding source

The funder of the study designed and oversaw the conduct of the trial. The investigators were responsible for participant-level data collection and worked under confidentiality agreements with the funder. The funder conducted site monitoring, study-level data collection and dataset creation, and data analysis.

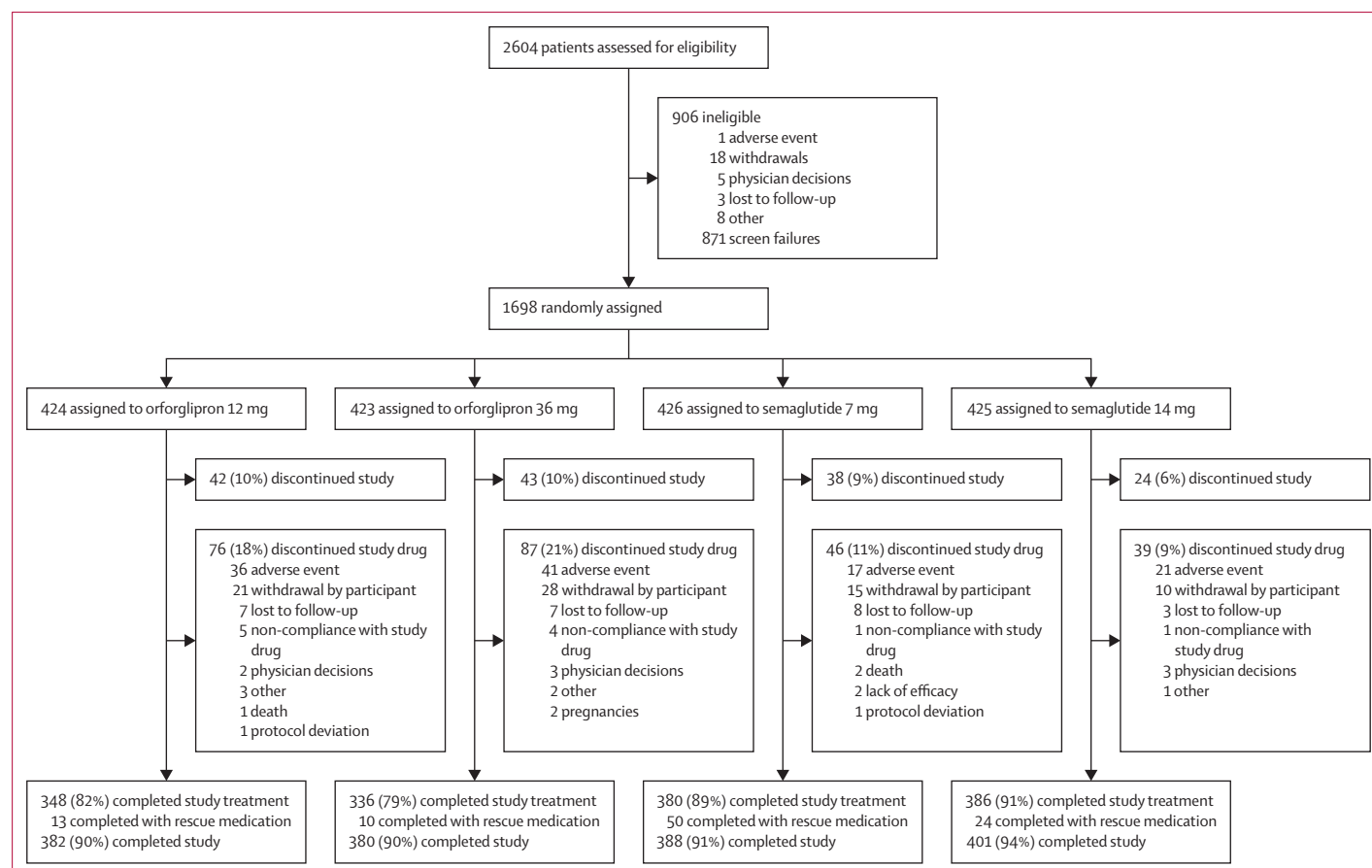


Figure 1: Participant flowchart

Results

Between Sept 22, 2023, and Aug 22, 2025, we screened 2604 participants and 1698 participants were randomly assigned to orforglipron 12 mg (n=424), orforglipron 36 mg (n=423), semaglutide 7 mg (n=426), or semaglutide 14 mg (n=425; figure 1). A total of 1551 (91%) participants completed the study. The demographics and clinical characteristics were similar between groups (table 1). The mean age of participants was 53·9 years (SD 11·1), 825 (48·6%) were female, 873 (51·4%) were male, mean baseline duration of diabetes was 8·6 years (SD 6·5), mean HbA_{1c} was 8·3% (SD 1·1), mean bodyweight was 97·0 kg (22·3), and mean BMI was 35·1 kg/m² (7·1). There was more treatment discontinuation in orforglipron groups (76 [18%] of 424 on 12 mg; 87 [21%] of 423 on 36 mg) than semaglutide groups (46 [11%] of 426 on 7 mg; 39 [9%] of 425 on 14 mg), with adverse events as the main reason (figure 1; appendix p 33). More participants in the orforglipron groups (36 [8%] on

12 mg; 41 [10%] on 36 mg) discontinued study treatment due to adverse events than in the semaglutide groups (17 [4%] on 7 mg; 21 [5%] on 14 mg; appendix p 33). Fewer participants completed study treatment with rescue therapy for severe persistent hyperglycaemia in orforglipron groups: 13 (3%) of 424 in the orforglipron 12 mg group and 10 (2%) of 423 in the orforglipron 36 mg group compared with 50 (12%) of 426 in the semaglutide 7 mg group and 24 (6%) of 425 in the semaglutide 14 mg group (figure 1).

Orforglipron met the primary objective of non-inferiority (margin 0·3%) and met the multiplicity-controlled key secondary objective of superiority to semaglutide in mean change from baseline in HbA_{1c} at week 52 (figure 2A; appendix pp 29–30). For the treatment regimen estimand, from a mean overall baseline HbA_{1c} of 8·3%, the mean change in HbA_{1c} at week 52 for orforglipron 12 mg was –1·71% (SE 0·07), for orforglipron 36 mg was –1·91% (0·08), for

	Orforglipron 12 mg (n=424)	Orforglipron 36 mg (n=423)	Semaglutide 7 mg (n=426)	Semaglutide 14 mg (n=425)	Overall (n=1698)
Age, years	54·0 (11·3)	53·4 (11·4)	54·9 (11·2)	53·4 (10·5)	53·9 (11·1)
Sex					
Female	195 (46%)	218 (52%)	206 (48%)	206 (48%)	825 (48·6%)
Male	229 (54%)	205 (48%)	220 (52%)	219 (52%)	873 (51·4%)
Race					
White	250/418 (60%)	259/415 (62%)	266/419 (63%)	254/420 (60%)	1029/1672 (61·5%)
American Indian or Alaska Native	90/418 (22%)	79/415 (19%)	76/419 (18%)	89/420 (21%)	334/1672 (20·0%)
Asian	50/418 (12%)	51/415 (12%)	52/419 (12%)	49/420 (12%)	202/1672 (12·1%)
Black or African American	17/418 (4%)	16/415 (4%)	12/419 (3%)	20/420 (5%)	65/1672 (3·9%)
Ethnicity					
Hispanic or Latino	304 (72%)	304 (72%)	316 (74%)	310 (73%)	1234 (72·7%)
Not Hispanic or Latino	117 (28%)	117 (28%)	108 (25%)	113 (27%)	455 (26·8%)
Diabetes duration, years	9·0 (7·1)	8·2 (5·8)	8·7 (6·3)	8·6 (6·9)	8·6 (6·5)
HbA _{1c} , concentration					
In %	8·3 (1·0)	8·3 (1·1)	8·3 (1·1)	8·3 (1·1)	8·3 (1·1)
In mmol/mol	66·8 (11·5)	67·0 (12·1)	67·3 (11·9)	67·2 (12·3)	67·1 (12·0)
≤8·0% [n (%)]	202 (48%)	217 (51%)	199 (47%)	200 (47%)	818 (48·2%)
>8·0% [n (%)]	222 (52%)	206 (49%)	227 (53%)	225 (53%)	880 (51·8%)
Fasting serum glucose					
In mg/dL	169 (54)	167 (52)	170 (51)	172 (52)	169 (52)
In mmol/L	9·4 (3·0)	9·3 (2·9)	9·4 (2·8)	9·6 (2·9)	9·4 (2·9)
BMI, kg/m ²	34·5 (7·1)	35·3 (7·2)	35·1 (7·0)	35·5 (7·1)	35·1 (7·1)
Weight, kg	96·1 (22·2)	96·5 (21·9)	97·1 (22·6)	98·4 (22·6)	97·0 (22·3)
Waist circumference, cm	111·4 (14·9)	112·3 (15·2)	112·3 (15·6)	113·4 (15·4)	112·3 (15·3)
Mean estimated glomerular filtration rate, mL per min per 1·73 m ²	90·4 (20·3)	92·4 (19·2)	88·9 (19·7)	90·5 (19·2)	90·6 (19·6)
Blood pressure, mm Hg					
Systolic	129·6 (13·8)	129·1 (13·1)	129·5 (13·4)	128·8 (13·7)	129·2 (13·5)
Diastolic	80·0 (9·2)	79·9 (8·3)	80·2 (9·1)	80·1 (8·9)	80·0 (8·9)
Pulse rate, bpm	74·7 (10·0)	75·1 (10·1)	74·8 (10·3)	76·2 (10·0)	75·2 (10·1)
Data are mean (SD) or n (%), unless otherwise indicated. This table includes all randomly assigned participants. Estimated glomerular filtration rate is calculated by use of the Chronic Kidney Disease Epidemiology cystatin-C equation.					
Table 1: Demographics and clinical characteristics of participants at baseline					

semaglutide 7 mg was -1.23% (0.05), and for semaglutide 14 mg was -1.47% (0.06 ; figure 2A; appendix pp 29–30). Both orforglipron doses were non-inferior and also superior to both doses of semaglutide, with estimated treatment differences ranging from -0.24 to -0.68 (figure 2A; appendix pp 29–30). Similar results and identical inferences were obtained for the efficacy estimand (figure 2B; appendix pp 31–32). Participants assigned to orforglipron groups had greater dose-related mean improvements in HbA_{1c} than those assigned to semaglutide groups, beginning at week 4 (the earliest timepoint measured) and continuing for 52 weeks (figure 2A, B). The final mean HbA_{1c} at end of treatment was 6.4% ($SE\ 0.06$) in the orforglipron 12 mg group, 6.1% (0.05) in the orforglipron 36 mg group, 7.2% (0.06) in the semaglutide 7 mg group, and 6.8% (0.06) in the semaglutide 14 mg group (efficacy estimand; appendix p 23).

Compared with semaglutide, both orforglipron groups had a significantly higher proportion of participants who reached HbA_{1c} target values of less than 7.0% (307 [72%] of 424 on orforglipron 12 mg and 321 [76%] of 423 on orforglipron 36 mg vs 228 [54%] of 426 on semaglutide 7 mg and 270 [64%] of 425 on semaglutide 14 mg), 6.5% or less (266 [63%] on orforglipron 12 mg and 287 [68%] on orforglipron 36 mg vs 162 [38%] on semaglutide 7 mg and 204 [48%] on semaglutide 14 mg), and less than 5.7% (91 [21%] on orforglipron 12 mg and 133 [31%] on orforglipron 36 mg vs 31 [7%] on semaglutide 7 mg and 50 [12%] on semaglutide 14 mg) at week 52 for treatment regimen estimand (figure 2C; appendix pp 29–30). Of those who had HbA_{1c} of less than 5.7% , all (100%) 91 participants in the orforglipron 12 mg group, 130 (98%) of 133 in the orforglipron 36 mg group, and 30 (97%) of 31 in the semaglutide 7 mg group, and all (100%) 50 in the semaglutide 14 mg group did so without level 2 or 3 hypoglycaemia. Similar significantly higher proportions of participants on orforglipron achieving HbA_{1c} targets were observed for the efficacy estimand (figure 2D).

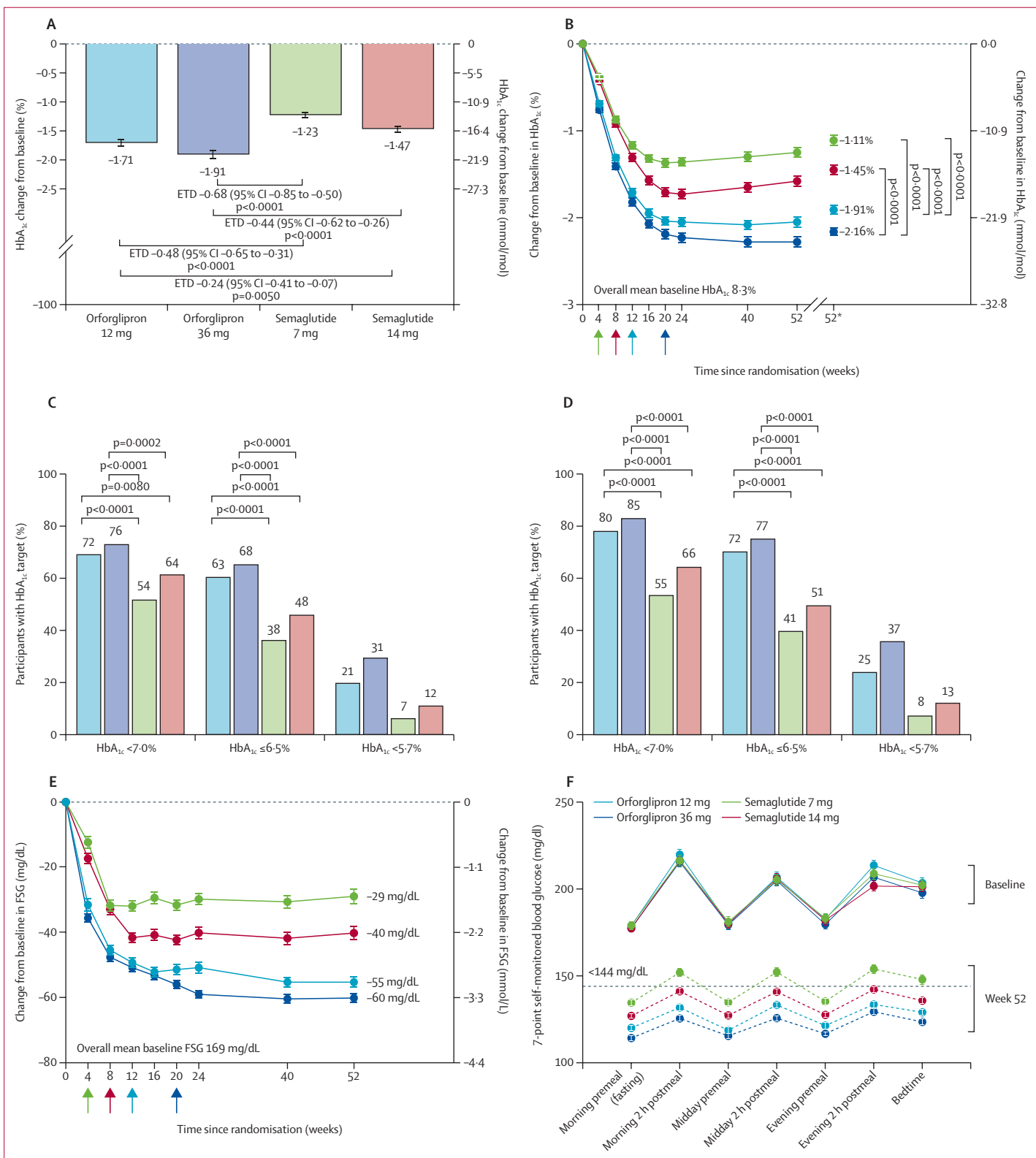
In a prespecified subgroup analysis, the change from baseline in HbA_{1c} favoured orforglipron compared with both doses of semaglutide in each baseline HbA_{1c} subgroup. Participants with baseline HbA_{1c} of more than 8.0% had larger reductions in HbA_{1c} (orforglipron -2.1% [$SE\ 0.1$] on 12 mg and -2.6% [0.1] on 36 mg vs semaglutide -1.7% [0.08] on 7 mg and -2.0% [0.09] on 14 mg) than those with a baseline HbA_{1c} of 8.0% or less (orforglipron -1.2% [$SE\ 0.06$] on 12 mg and -1.2% [0.1] on 36 mg vs semaglutide 0.7% [0.07] on 7 mg and -0.9% [0.07] on 14 mg; appendix p 24).

Fasting serum glucose decreased from baseline to week 52 with greater reductions observed with orforglipron than semaglutide beginning at week 4 and continuing to week 52 (efficacy estimand; figure 2E). Based on participant self-monitored blood glucose profiles, treatment with both orforglipron doses was associated with a greater reduction from baseline in

overall mean daily, pre-meal, and 2-h post-meal glucose values at week 52 compared with both semaglutide doses, with both doses of orforglipron maintaining the mean glucose concentrations below 144 mg/dL (8 mmol/L)²¹ at every timepoint measured throughout the day (efficacy estimand; figure 2F).

Both orforglipron groups had significant weight reductions from baseline, starting from week 4 and continuing to week 52 (figure 3A, B). For the treatment regimen estimand, from a mean overall baseline bodyweight of 97.0 kg , mean percentage bodyweight changes to week 52 were -6.1% ($SE\ 0.4$) with orforglipron 12 mg, -8.2% (0.4) with orforglipron 36 mg, -3.9% (0.3) with semaglutide 7 mg, and -5.3% (0.3) with semaglutide 14 mg (figure 3A; appendix pp 29–30). With the exception of orforglipron 12 mg versus semaglutide 14 mg, for which the comparison was not significant (estimated treatment difference -0.8 ; $95\% CI -1.7$ to 0.2), orforglipron treatment was superior to semaglutide for weight loss with estimated treatment differences of -2.3% ($95\% CI -3.2$ to -1.3 ; $p<0.0001$) with orforglipron 12 mg versus semaglutide 7 mg, -4.3% (-5.3 to -3.3 ; $p<0.0001$) with orforglipron 36 mg versus semaglutide 7 mg, and -2.8% (-3.9 to -1.9 ; $p<0.0001$) with orforglipron 36 mg versus semaglutide 14 mg (figure 3A; appendix pp 29–30). Results for mean change from baseline in bodyweight (kg) are presented in the appendix (pp 25, 29–30). Both orforglipron doses were superior to both doses of semaglutide in terms of mean percent change in bodyweight for the efficacy estimand (figure 3B; appendix pp 31–32). Mean reductions in bodyweight and percentage bodyweight with orforglipron were dose dependent (figure 3B; appendix p 25). Greater proportions of participants treated with orforglipron than those treated with semaglutide had reductions in bodyweight of 5% or more (orforglipron 250 [59%] on 12 mg and 295 [69%] on 36 mg vs semaglutide 157 [37%] on 7 mg and 208 [49%] on 14 mg), 10% or more (orforglipron 119 [28%] on 12 mg and 185 [44%] on 36 mg vs semaglutide 56 [13%] on 7 mg and 88 [21%] on 14 mg), and 15% or more (orforglipron 51 [12%] on 12 mg and 100 [23%] on 36 mg vs semaglutide 21 [5%] on 7 mg and 27 [6%] on 14 mg; efficacy estimand; figure 3C; appendix 31–32). Mean reduction in waist circumference and BMI was larger in participants treated with orforglipron 36 mg than all other groups (figure 3D; appendix p 26).

At week 52, for the efficacy estimand, there were improvements from baseline in non-HDL cholesterol, triglycerides, VLDL cholesterol, total cholesterol, and HDL cholesterol in most treatment groups (figure 3E; appendix p 34). Mean systolic blood pressure decreased over the course of the study in all groups, with greater decreases with orforglipron 12 mg and 36 mg than with semaglutide 7 mg and 14 mg (figure 3F). No clinically relevant differences were observed in mean change from baseline in diastolic blood pressure across the



orforglipron groups and semaglutide groups (appendix p 28).

Adverse events were reported by 317 (75%) of 424 participants receiving orforglipron 12 mg, 318 (75%) of 423 participants receiving orforglipron 36 mg, 303 (71%) of 426 participants receiving semaglutide 7 mg, and 308 (72%) of 425 participants receiving semaglutide 14 mg (table 2). Permanent treatment discontinuations due to adverse events occurred in 37 (9%) of participants receiving orforglipron 12 mg, 41 (10%) receiving orforglipron 36 mg, 19 (4%) receiving semaglutide 7 mg, and 21 (5%) receiving semaglutide 14 mg (table 2).

The frequency of serious adverse events was similar between the orforglipron 12 mg and both semaglutide groups (table 2; appendix p 35). Although the overall frequency of serious adverse events was higher in the orforglipron 36 mg group than the other groups, about half of the participants had these serious events before escalating to the 36 mg dose (table 2; appendix p 35). There were four deaths overall: three deaths were adjudicated to a cardiovascular cause (one in the orforglipron 12 mg group and two in semaglutide 7 mg group) and one death in the orforglipron 36 mg group was adjudicated to non-cardiovascular cause (a malignancy; appendix p 36).

The most frequent adverse events reported were gastrointestinal-related and occurred more frequently with orforglipron than semaglutide (table 2). Most reports of nausea, diarrhoea, and vomiting in the orforglipron groups were mild to moderate in severity (appendix p 37), occurred early in the study during the dose escalation period, and diminished over time (appendix p 27). The proportion of participants who

discontinued the study drug due to gastrointestinal adverse events was higher in orforglipron groups than semaglutide groups (table 2).

Clinically significant (level 2) hypoglycaemia (blood glucose <54 mg/dL [<3.0 mmol/L]) was reported in three (1%) participants receiving orforglipron 12 mg, four (1%) receiving orforglipron 36 mg, two ($<1\%$) receiving semaglutide 7 mg, and one ($<1\%$) receiving semaglutide 14 mg; one case of severe hypoglycaemia was reported in a participant receiving orforglipron 36 mg (table 2). This participant reported skipping meals over the 3-day period preceding the hypoglycaemia event. The participant continued study treatment and did not report any subsequent level 2 or severe hypoglycaemia events. No participant discontinued the study drug due to hypoglycaemia. More participants assigned to semaglutide than to orforglipron required rescue therapy for persistent severe hyperglycaemia during the study (table 2).

Four cases of adjudication-confirmed acute pancreatitis were reported: two in the orforglipron 12 mg group and two in the semaglutide 14 mg group (table 2). No cases of pancreatic cancer were reported.

No cases of medullary thyroid cancer were reported, and papillary thyroid cancer was reported in a participant in the orforglipron 12 mg group ($<1\%$) and a participant in the semaglutide 14 mg group ($<1\%$).

Mean pulse rate increased in both participants assigned orforglipron and those assigned semaglutide (appendix p 28). At week 52, mean increases in pulse rate were greater in the orforglipron groups (12 mg 3.7 bpm; 36 mg 4.7 bpm) than the semaglutide groups (7 mg 1.0 bpm; 14 mg 1.5 bpm). Despite higher mean increases from baseline in pulse rate, the proportion of people with treatment-emergent tachycardia (pulse rate >100 bpm) or persistent tachycardia (pulse rate >100 bpm at two or more consecutive visits) was similar between orforglipron and semaglutide, as were the incidences of adverse events related to supraventricular tachyarrhythmias and tachycardias (appendix p 38). A total of 19 (1.1%) of all 1698 participants had an adjudication-confirmed major adverse cardiovascular event and incidence was similar across treatment groups (table 2).

Mean estimated glomerular filtration rate decreased throughout the 52 weeks in all groups (-3.1 to -5.4 mL per min per 1.73 m 2) with no clinically relevant differences between the orforglipron and semaglutide groups (appendix p 39). Urine albumin to creatinine ratio also decreased in all groups, with greater percentage reductions with orforglipron (-31.4% [SE 3.0] on 12 mg and -32.5% [3.1] on 36 mg) versus semaglutide (-14.5% [4.2] on 7 mg and -23.8% [3.5] in 14 mg) at week 52 (appendix p 39).

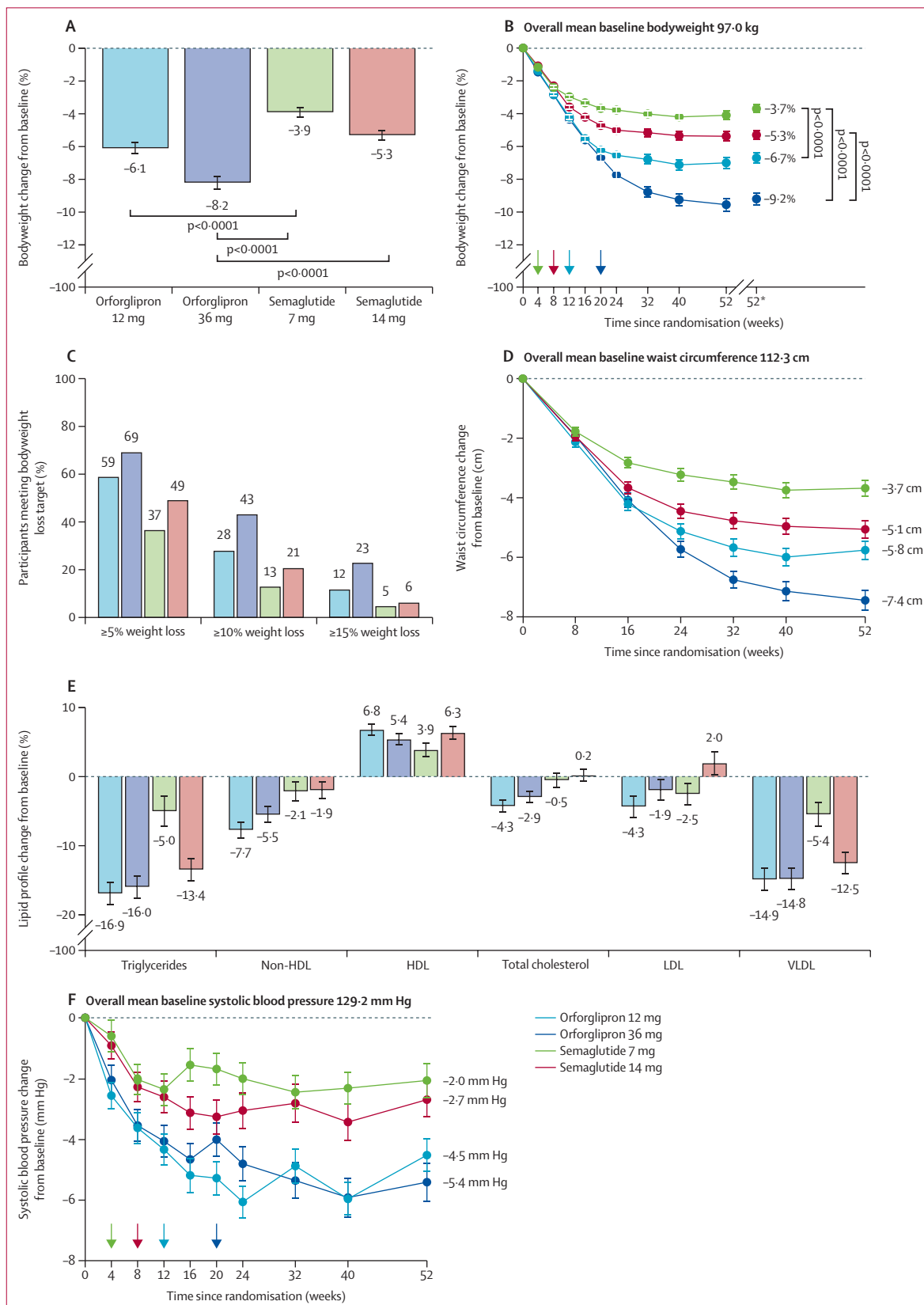
Based on evaluation of fundus photographs collected at baseline, 234 (14%) of 1696 participants assessed had diabetic retinopathy at baseline, primarily mild non-proliferative retinopathy (178 [10%] of 1696; appendix

Figure 2: HbA_{1c} changes, proportion of participants meeting HbA_{1c} targets, FSG, and 7-point self-monitored blood glucose profiles

(A) Change from baseline in HbA_{1c} at 52 weeks from ANCOVA with multiple imputation by treatment for missing HbA_{1c} data at 52 weeks (treatment regimen estimand). (B) Change from baseline in HbA_{1c} over time. Data are observed mean (SE), with MBE (SE) and MMRM analysis (efficacy estimand) shown in the side panel. (C) Proportion of participants meeting HbA_{1c} targets $<7.0\%$, $\leq 6.5\%$, and $<5.7\%$ at 52 weeks (treatment regimen estimand) with p values for risk difference; the proportion of participants with HbA_{1c} $<5.7\%$ was not controlled for type 1 errors; the proportion was calculated with imputed data from a logistic regression model with primary multiple imputation. (D) Proportion of participants with HbA_{1c} targets $<7.0\%$, $\leq 6.5\%$, and $<5.7\%$ at 52 weeks (efficacy estimand) with p values for risk difference; the proportion of participants with HbA_{1c} $<5.7\%$ was not controlled for type 1 errors; the proportion was calculated with imputed data from logistic regression with multiple imputation under a missing at random assumption. (E) Change from baseline in FSG over time from MMRM analysis (efficacy estimand); this endpoint was not controlled for type 1 error. (F) 7-point self-monitored blood glucose profiles at baseline and week 52 from MMRM (efficacy estimand). Data are MBE (SE), unless otherwise noted, at 52 weeks for all randomly assigned participants. The primary and key secondary objectives were tested under a multiplicity-controlled procedure at a two-sided α level of 0.05 and p values presented are unadjusted two-sided for superiority. Arrows indicate when the randomly assigned dose of orforglipron 12 mg or 36 mg and semaglutide 7 mg or 14 mg were achieved (end of lead-in period). FSG=fasting serum glucose. ETD=estimated treatment differences. MBE=model-based estimate. MMRM=mixed model repeated measures. *MBE (SE).

Figure 3: Bodyweight changes, proportion of participants achieving weight loss targets, lipid profiles, and systolic blood pressure changes

(A) Percentage change from baseline in bodyweight at 52 weeks from ANCOVA with multiple imputation by treatment for missing weight at 52 weeks (treatment regimen estimand); orforglipron 12 mg versus semaglutide 14 mg was not controlled for type 1 errors. (B) Percentage change from baseline in bodyweight over time; data are observed mean (SE), with MBE (SE) and MMRM analysis (efficacy estimand) shown in the side panel; orforglipron 12 mg versus semaglutide 14 mg was not controlled for type 1 error. (C) Proportion of participants with weight loss $\geq 5\%$, $\geq 10\%$, and $\geq 15\%$ (efficacy estimand); these endpoints were not controlled for type 1 errors; the proportion was calculated with imputed data from logistic regression with multiple imputation under a missing-at-random assumption. (D) Change from baseline in waist circumference over time from MMRM (efficacy estimand); this endpoint was not controlled for type 1 errors. (E) Percentage change from baseline in lipids at 52 weeks; data are estimated percentage means (SE) with log transformation; these endpoints were not controlled for type 1 errors. (F) Change from baseline in systolic blood pressure over time from MMRM (efficacy estimand); this endpoint was not controlled for type 1 errors. Data are MBE (SE), unless otherwise noted, at 52 weeks for all randomly assigned participants. Arrows indicate when the randomly assigned dose of orforglipron 12 mg or 36 mg and semaglutide 7 mg or 14 mg were achieved. The primary and key secondary objectives were tested under a multiplicity-controlled procedure at a two-sided α level of 0.05 and p values presented are unadjusted two-sided for superiority. MBE=model-based estimate. MMRM=mixed model repeated measures. *MBE (SE).



p 40). The proportion of participants with worsening in diabetic retinopathy category on fundus photographs was similar across orforglipron and semaglutide groups in the total population (23 [6%] of 358 on orforglipron 12 mg, 27 [8%] of 360 on orforglipron 36 mg, 26 [7%] of 381 on semaglutide 14 mg, and 29 [8%] of 371 on semaglutide 7 mg) and in the subset with diabetic retinopathy at baseline (nine [18%] of 51 on orforglipron 12 mg, ten [18%] of 55 on orforglipron 36 mg, seven [13%] of 53 on semaglutide 14 mg, and 12 [24%] of 49 on

semaglutide 7 mg; appendix p 40). Overall, 84 (4.9%) of 1698 participants had an adverse event of diabetic retinopathy or a potentially related complication, with a similar frequency across the orforglipron and semaglutide groups (table 2). None of the adverse events of diabetic retinopathy resulted in visual loss and no ophthalmological interventions were reported related to these events.

Mean reductions in ALT and AST from baseline were observed across all groups at week 52 (appendix p 39).

	Orforglipron 12 mg (n=424)	Orforglipron 36 mg (n=423)	Semaglutide 7 mg (n=426)	Semaglutide 14 mg (n=425)	Overall (n=1698)
Participants with ≥1 treatment-emergent adverse event	317 (75%)	318 (75%)	303 (71%)	308 (72%)	1246 (73.4%)
Serious adverse events	21 (5%)	40 (9%)	19 (4%)	24 (6%)	104 (6.1%)
Deaths	1 (<1%)	1 (<1%)	2 (<1%)	0	4 (0.2%)
Adverse event leading to study treatment discontinuation	37 (9%)	41 (10%)	19 (4%)	21 (5%)	118 (6.9%)
Gastrointestinal adverse events leading to study treatment discontinuation	26 (6%)	25 (6%)	11 (3%)	13 (3%)	75 (4.4%)
Nausea	6 (1%)	5 (1%)	3 (1%)	4 (1%)	18 (1.1%)
Vomiting	6 (1%)	9 (2%)	3 (1%)	4 (1%)	22 (1.3%)
Diarrhoea	2 (<1%)	5 (1%)	4 (1%)	0	11 (0.6%)
Abdominal pain	2 (<1%)	3 (1%)	0	0	5 (0.3%)
Dyspepsia	5 (1%)	1 (<1%)	0	0	6 (0.4%)
Decreased appetite	0	3 (1%)	1 (<1%)	0	4 (0.2%)
Constipation	0	0	0	1 (<1%)	1 (0.1%)
Treatment-emergent adverse events occurring in ≥5% of participants in any treatment group by preferred term					
Nausea	111 (26%)	114 (27%)	54 (13%)	90 (21%)	369 (21.7%)
Diarrhoea	91 (21%)	97 (23%)	55 (13%)	76 (18%)	319 (18.8%)
Vomiting	72 (17%)	70 (17%)	32 (8%)	44 (10%)	218 (12.8%)
Dyspepsia	47 (11%)	43 (10%)	18 (4%)	33 (8%)	141 (8.3%)
Decreased appetite	30 (7%)	29 (7%)	21 (5%)	27 (6%)	107 (6.3%)
Hyperglycaemia	25 (6%)	20 (5%)	68 (16%)	32 (8%)	145 (8.5%)
Eructation	42 (10%)	31 (7%)	17 (4%)	14 (3%)	104 (6.1%)
Abdominal distension	24 (6%)	34 (8%)	11 (3%)	14 (3%)	83 (4.9%)
Gastro-oesophageal reflux disease	23 (5%)	24 (6%)	12 (3%)	18 (4%)	77 (4.5%)
Constipation	56 (13%)	51 (12%)	32 (8%)	38 (9%)	177 (10.4%)
Abdominal pain	21 (5%)	25 (6%)	13 (3%)	14 (3%)	73 (4.3%)
Hypertension	10 (2%)	10 (2%)	14 (3%)	22 (5%)	56 (3.3%)
Flatulence	22 (5%)	12 (3%)	14 (3%)	7 (2%)	55 (3.2%)
All gastrointestinal adverse events	249 (59%)	245 (58%)	157 (37%)	193 (45%)	844 (49.7%)
Other adverse events of interest					
Hypoglycaemia (blood glucose <54 mg/dL)*	3 (1%)	4 (1%)	2 (<1%)	1 (<1%)	10 (0.6%)
Severe hypoglycaemia*	0	1 (<1%)	0	0	1 (0.1%)
Severe persistent hyperglycaemia requiring rescue therapy	15 (4%)	12 (3%)	56 (13%)	25 (6%)	108 (6.4%)
Adjudication-confirmed pancreatitis	2 (<1%)	0	0	2 (<1%)	4 (0.2%)
Thyroid malignancies†	1 (<1%)	0	0	1 (<1%)	2 (0.1%)
Gallbladder disorders	5 (1%)	6 (1%)	3 (1%)	4 (1%)	18 (1.1%)
Cholelithiasis	4 (1%)	6 (1%)	3 (1%)	3 (1%)	16 (0.9%)
Diabetic retinopathy, diabetic macular oedema, and related complications‡	21 (5%)	18 (4%)	26 (6%)	19 (4%)	84 (4.9%)
Adjudication-confirmed major adverse cardiovascular events	4 (1%)	4 (1%)	8 (2%)	3 (1%)	19 (1.1%)

Data are n (%) for safety participants (all participants who received at least one dose of the study drug). *Excluding hypoglycaemic events occurring after initiation of any new antihyperglycaemic therapy. †No cases of medullary thyroid cancer were reported, and papillary thyroid cancer was reported in one (<1%) participant in the orforglipron 12 mg group and one (<1%) participant in the semaglutide 14 mg group. ‡Based on MedDRA Preferred Term search defined in appendix (p 40).

Table 2: Adverse events in the safety population

The percentage of participants with ALT or AST at least 3 times, at least 5 times, or at least 10 times the upper limit of normal at any point after baseline was generally similar between orforglipron and semaglutide groups (appendix p 41). No-one met Hy's Law criteria for hepatocellular drug-induced liver injury.

Improvements from baseline to week 52 were seen in all Short Form-36 Health Survey domain and component summary scores in all treatment groups (efficacy estimand; appendix p 43).

Discussion

ACHIEVE-3 is the first study to directly compare the efficacy and safety of two oral GLP-1 receptor agonists in a randomised controlled trial. It showed that both 12 mg and 36 mg orforglipron doses were superior to both 7 mg and 14 mg semaglutide doses in reducing HbA_{1c} and meeting HbA_{1c} targets of less than 7·0% and 6·5% or less at week 52, in a dose-dependent manner. The difference in glycaemic control was already significant at week 4 and was sustained until the end of the trial (week 52) for orforglipron. In a prespecified analysis for participants with higher baseline HbA_{1c} (>8·0%), orforglipron further showed particularly robust efficacy with reductions of up to 2·6%, compared with 2·0% for those receiving semaglutide.

In terms of therapeutic targets, 72–76% of participants on orforglipron and 54–64% of participants on semaglutide met the recommended HbA_{1c} level of less than 7·0%.²² In ACHIEVE-3, only 7–12% of participants receiving semaglutide developed near normoglycaemia (HbA_{1c} <5·7%) compared with 21–31% with orforglipron, of these, 98–100% did not have level 2 or 3 hypoglycaemia. These data show that stricter glycaemic targets, beyond 6·5%, could now be feasibly and safely achieved with oral therapy.

Weight reduction can lead to clinical and metabolic improvements and diabetes reversion to normoglycaemia, in addition to reducing the risk of weight-related complications.²² Participants receiving orforglipron had significantly greater reductions in bodyweight than those receiving semaglutide, which also translated into the proportion of participants meeting clinically meaningful weight reduction thresholds. Weight reduction in the range of 5–15% has been recommended as the primary target of management for many people living with type 2 diabetes, with 5–10% weight reduction conferring metabolic improvements and more than 10% weight reduction having disease-modifying effects, including potential remission of diabetes and reduction in cardiovascular risk.^{23,24} There was a weight reduction of 5% or more in 59–69% of participants on orforglipron and 37–49% of participants on semaglutide, and 28–43% of participants on orforglipron and 13–21% of participants on semaglutide had a weight reduction of 10% or more.

Oral semaglutide has shown cardiovascular risk reduction in the SOUL trial.¹⁶ Orforglipron, by reducing

waist circumference and improving systolic blood pressure and lipid measures to a greater extent than semaglutide, merits further investigation and these results might suggest a role for orforglipron in managing cardiometabolic risk factors. A cardiovascular outcomes study of orforglipron is currently underway (NCT07241390) and will identify whether these effects, in combination with HbA_{1c} and weight reductions will translate to improved cardiovascular outcomes.

The safety profile of orforglipron and semaglutide in this trial was generally consistent with the GLP-1 receptor agonist class in people with type 2 diabetes,^{25,26} although there were some slight differences regarding gastrointestinal events and pulse rate. The most common adverse events were mostly mild-to-moderate gastrointestinal symptoms. Gastrointestinal events were reported more frequently in the orforglipron groups than the semaglutide groups, and a higher proportion of participants in the orforglipron groups discontinued treatment due to these events. Increases from baseline in pulse and heart rate is a well established class effect of GLP-1 receptor agonists but were greater with orforglipron (mean increase of 4–5 bpm) than semaglutide (mean increase of 1–2 bpm) in this study. Selection between oral options for GLP-1 receptor agonists could ultimately be guided by prioritisation of glycaemic control, weight reduction, and proven cardiovascular disease risk reduction, balanced against individual differences in tolerability (including sensitivity to gastrointestinal symptoms), administration convenience, and other patient-specific factors.

Serious adverse events were balanced between orforglipron 12 mg (5%) and semaglutide 7 mg (4%) and 14 mg (6%) groups; however, they were slightly higher in the orforglipron 36 mg group (9%), although most serious adverse events in this group occurred before receiving maintenance dose. Treatment discontinuation rates due to gastrointestinal adverse events in the orforglipron groups were higher than the semaglutide groups, and consistent with previously completed phase 3 orforglipron trials in people with obesity or overweight without type 2 diabetes.²⁷ Discontinuation rates due to gastrointestinal adverse events in the semaglutide groups were lower than equivalent doses in previous semaglutide phase 3 trials.^{26,28} Adverse events related in general to supraventricular tachyarrhythmias or specifically tachycardia were infrequent and balanced across treatment groups.

Other safety outcomes of relevance to the GLP-1 receptor agonist class were balanced across treatment groups, including the occurrence of adjudication-confirmed pancreatitis and major adverse cardiovascular events, thyroid malignancies, gallbladder disorders, and changes in diabetic retinopathy. Hepatic safety was closely monitored and was similar across the two drugs.

Strengths of this trial include a large and diverse population across multiple countries and consistency of

results. The reductions in HbA_{1c} concentration and bodyweight observed with orforglipron were consistent with the results already reported in ACHIEVE-1¹⁹ and ATTAIN-2,²⁵ while the glycaemic efficacy of semaglutide was consistent with the PIONEER phase 3 trials, indicating that trial conditions and execution were appropriate to detect treatment differences in this comparison.

Limitations include the open-label design, despite being masked to orforglipron doses, which could have introduced bias in subjective assessments, such as patient-reported outcomes. Although primary and key secondary efficacy endpoints (HbA_{1c} and weight) are objective measures less susceptible to bias, the awareness of treatment assignment could still influence patient behaviour and adverse event reporting. Other limitations include that self-reported gastrointestinal adverse events represent a potential source of bias, although this reporting method is standard in most clinical trials, and that a treatment duration beyond 52 weeks could be designed to collect longer-term safety data.

In conclusion, oral orforglipron was non-inferior and superior to semaglutide in terms of efficacy, with meaningful improvements in glycaemic control and weight reduction compared with oral semaglutide in patients with type 2 diabetes, and with larger improvements in cardiometabolic risk factors and simplified administration. The overall safety profile of both orforglipron and semaglutide were consistent with the GLP-1 receptor agonist class, although patients receiving orforglipron had a higher incidence of gastrointestinal events and larger mean increases in pulse rate than those receiving oral semaglutide. Orforglipron represents a potential new therapeutic option for individuals with type 2 diabetes considering initiation of GLP-1 receptor agonist therapy who might prefer an alternative to the subcutaneous route of administration.

Contributors

The funder (Eli Lilly) conceived, designed, and supervised the trial, which included site monitoring, data collection, and analysis. YZ, DC, W-SW, and MD contributed to the study design. YZ and MD provided medical oversight during the trial. JL and RL were responsible for the data curation, formal analysis, validation, visualisation, and editing of statistical analyses. JR and YZ wrote the first draft of the manuscript. JR, DY, YZ, DC, W-SW, RL, and MD drafted the manuscript. All authors participated in the interpretation of the data and critical review of the manuscript, approved the final and previous versions of the manuscript, had full access to all the data in the study, and had final responsibility for the decision to submit for publication.

Declaration of interests

JL, MD, W-SW, RL, DC, and YZ are current employees and shareholders of Eli Lilly. JR has served on scientific advisory boards and received honoraria or consulting fees from Amgen, Applied Therapeutics, Biomea Fusion, Boehringer Ingelheim, Corcept, Ecogene, Eli Lilly, Novo Nordisk, Oramed Pharmaceuticals, Regeneron, Regor, Sanofi, Scholar Rock, Structure Therapeutics, and Terns Pharmaceuticals and has received grants and research support from Amgen, Applied Therapeutics, Biomea Fusion, Boehringer Ingelheim, Corcept, Eli Lilly, Hanmi Pharmaceutical, Merck, Novo Nordisk, Oramed Pharmaceuticals, Pfizer, Regeneron, Regor, Sanofi, Structure Therapeutics, and Terns Pharmaceuticals. DY reports support from Eli Lilly for the present manuscript; grants or contracts from Boehringer Ingelheim, Taisho

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Data sharing

Eli Lilly provides access to all individual participant data collected during the trial, after anonymisation, with the exception of pharmacokinetic or genetic data. Data are available to request 6 months after the indication studied has been approved in the USA and EU or after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available. Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data sharing agreement. Data and documents, including the study protocol, statistical analysis plan, clinical study report, and blank or annotated case report forms, will be provided in a secure data sharing environment. For details on submitting a request, see the instructions provided at www.vivli.org.

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